ABHISHEK KUMAR

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Overview

Computer Science undergraduate with strong foundations in **Data Structures, Algorithms, Software Engineering, System Design, and Backend Development**. Experienced in designing and building scalable full-stack and backend applications using **React, Next.js, Node.js, TypeScript, REST APIs, SQL databases, and event streaming pipelines**, with practical focus on **clean architecture, problem solving, and maintainable code**.

Technical Skills

Problem Solving & CS Fundamentals: Data Structures, Algorithms, Problem Solving, OOP, DBMS, OS, Computer Networks
Frontend Architecture: React.js, Next.js 16 (App Router), Tailwind CSS, Framer Motion, Axios, State Management
Backend Development: Node.js, Express.js, REST APIs, JWT Session Rotation, RBAC, Redis Queuing, Rate Limiting
Databases & ORM: PostgreSQL (Neon Serverless), MongoDB (Mongoose), Redis (Upstash), Prisma ORM, Normalization
Languages & Tools: JavaScript (ES6+), TypeScript, SQL, HTML5, CSS3, Git, GitHub, Postman
DevOps, Cloud & AI Tooling: Google Gemini AI API, Docker, Vercel, Render, Polar.sh SDK, Zod Validation

Technical Projects & Experience

PulseEngine | Live Demo | GitHub **2026**

- **Tools:** Next.js 16, TypeScript, Upstash Redis, Neon PostgreSQL, Prisma, Google Gemini AI, Docker, Polar.sh
- Architected a distributed telemetry ingestion pipeline processing event streams with **sub-3ms latency**, leveraging an in-memory Redis buffer (`LPUSH`) to prevent database connection pool exhaustion under high concurrency.
- Engineered an asynchronous worker daemon handling micro-batch persistence (**500 events/batch**) into Neon Serverless PostgreSQL via Prisma ORM, scaling throughput up to **100k+ events/sec**.
- Integrated **Google Gemini 3.8 Flash** to build an autonomous AI SRE Copilot executing real-time root-cause analysis (RCA), blast radius evaluation, and generating copyable 1-click remediation shell commands upon SLA breaches.
- Implemented **Polar.sh subscription billing** with automated webhook synchronization, tenant quota metering, and dynamic soft-cap rate limiting; containerized the engine with multi-stage Docker builds.

CareConnect | Live Demo | GitHub **2025 - 2026**

- **Tools:** React.js, Node.js, Express.js, MongoDB, Tailwind CSS, Zod, JWT, Cloudinary, Vercel, Render
- Architected a 3-tier MERN platform (**Controller -> Service -> Repository**) serving 13 modular REST endpoints for doctor discovery, atomic appointment scheduling, and patient management.
- Engineered an **Axios response interceptor queue pipeline** that intercepts 401 statuses, coordinates token rotation, and replays enqueued requests, eliminating **100% of concurrent session drops**.
- Implemented atomic MongoDB appointment slot reservations preventing concurrent double-booking, enforced Role-Based Access Control (RBAC) with HTTP-only JWTs, and sanitized inputs via Zod schemas.
- Integrated multipart media uploads via Multer and Cloudinary SDK; streamlined diagnostics using Winston and Morgan, deploying backend APIs to Render.

Interactive Developer Portfolio | Live Demo | GitHub **2026**

- **Tools:** React 19, Tailwind CSS v4, Framer Motion, Web Audio API, Vite, Vercel
- Engineered an interactive portfolio achieving a **98+ Google Lighthouse performance score**, featuring interactive 3D physics cards, dynamic theme customization, and zero-asset browser Web Audio API sound synthesis.
- Implemented instant zero-JS obsidian preloading and `<link rel="preload">` asset pipelines with skeleton placeholders, achieving **< 150ms** first contentful paint (FCP).

Education

Bachelor of Computer Applications (BCA) **2023 - 2026**

DAV Centenary College, Faridabad | Affiliated to **Maharshi Dayanand University (MDU), Rohtak** | **Aggregate: 71.03%**
Relevant Coursework: Data Structures & Algorithms, Database Management Systems (SQL), Operating Systems, Computer Networks, Software Engineering, Object-Oriented Programming (Java/C++).

Senior Secondary (Class XII) - Science Stream **HBSE Board | Score: 77.2%**

Govt. Model Sanskriti Senior Secondary School (GMSSSS)

Secondary (Class X) - Matriculation **HBSE Board | Score: 92.8%**

Govt. Model Sanskriti Senior Secondary School (GMSSSS)

Certifications & Algorithmic Credentials

- **LeetCode Algorithmic Milestones:** Completed **130+ algorithmic challenges in JavaScript & TypeScript** (**Profile Link**) across Arrays, Two Pointers, Sliding Window, Graphs, and DP Optimization.
- **Full Stack Web Development 1.0:** Physics Wallah Skills (**Credential Link**) — Comprehensive curriculum in React.js, Node.js, Express, MongoDB, and REST API Architecture (Dec 2024).
- **Namaste DSA:** NamasteDev by Akshay Saini (**Credential Link**) — Rigorous training on Data Structures, Algorithms, Time-Space Complexity Analysis, and algorithmic design patterns.